

Karate Kata Style Classification Using Pose Landmarks and Deep Learning

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Abstract—This paper aims to shed light on the efficiency of using LSTM in the binary classification of Karate Kata players' styles. In addition, this paper showcases the results of the proposed LSTM model on the MPII Human Pose Dataset. Since our focus in this research is LSTMs, our paper also showcases the different approaches we applied in order to reach the best classification results. Moreover, the approach we settled on was to classify each video as a whole, using a player's landmarks as input for the LSTM model. This study concludes that in comparison to segmenting the video and classifying the moves individually, the approach of classifying the video as a whole showed significant improvement. It was also concluded that using LSTMs was efficient in the task of classifying the Karate Kata style of the player with an accuracy of 90.3%. Furthermore, the model's precision, recall, and F1-Score were 91%, 90%, and 90% respectively. It is also worth mentioning that different approaches such as using classical machine learning models also showed promising results. Further research in the field is needed in order to reach a more concise result.

Index Terms—Human-Computer Interaction, Deep Learning, LSTMs, Pose Landmarks, Karate Kata

I. INTRODUCTION

Karate Kata is a form of martial arts that involves performing a sequence of Karate movements with certain defensive techniques that require both physical and mental discipline. Even though Kata has been practiced for many centuries, instructors and students may still encounter some challenges. For instance, traditional evaluation methods can be subjective to each judge, leading to inconsistent judgment and feedback. Trainers may also find it challenging to give each student detailed, individualized feedback, specifically in big training groups. It is also challenging to classify Karate Kata techniques due to the rapidity of the body movements [1]. Previous research has used a variety of machine learning algorithms to classify various features of Karate movements. For instance, earlier research [2], [3], [4], [5] have used deep neural networks, particularly Convolutional Neural Networks (CNNs). In other research [6], classical machine learning models such as support vector machines (SVM) were used. Moreover, research on Kata performance has concentrated on evaluating specific movements as either good or bad or by classifying certain

Karate movements [7], [8], [9], [10], [11]. Although previous research has used deep neural networks and classical machine learning models, the classification of Karate movements can cause some technical difficulties. Because of the variability and complexity of Karate moves, it is challenging to create reliable and robust algorithms capable of reliably classifying and recognizing various types of movements. Furthermore, videos of Karate performances can differ in terms of camera angle, lighting conditions, and background noise. This paper seeks to demonstrate the overall correctness of performance for all 20 Kata movements by classifying each video frame. Fig. 1 shows a frame of the video representing a bad player style of the first Kata movement, whereas Fig. 2 shows the video frame for a good player style of the same movement. After that, we provide players with an overall classification based on their performance. In return, this will make the evaluation easier, more accurate, and more efficient, and therefore provide better feedback to students. Furthermore, our study investigates the efficiency of using Deep Neural Networks in the task of classifying the style of a Karate Kata player. This paper analyzes the efficiency of Long Short-Term Memory neural networks or LSTMs in the task of binary classification that classifies whether a player has a good or bad Karate Kata style. This paper aims to shed light on the results of using



Fig. 1. Bad style of the first Kata movement



Fig. 2. Good style of the first Kata movement

Deep Neural Networks and LSTMs in different fields, as well as comparing them to other classification approaches.

II. RELATED WORK

Emad et al., [11] proposed an idea that used an infrared camera to record the real-time movements of Karate players throughout the first seven movements of Kata. They proposed a system that would track players' movements in real-time and generate feedback on the performance of players. The segmentation method used in this study was based on the pauses between each movement. The segmentation method was implemented by calculating the mean of each pair of consecutive windows and determining whether the difference is less than a predetermined threshold by the ISODATA algorithm, which denotes that the player is not moving. The F-DTW (Fast Dynamic Time Wrapping) was used to classify the data, achieving an accuracy of 91.07%. They have also proposed another method of an improved version of their paper in which they have presented a system that captures the Karate moves using kinetic v2 sensors [9]. The features extracted are the body joints of the player. This paper focuses on three Karate moves Age-Uke, Mae-Geri, Gedan-Barai, and Soto-Uke. The system works by capturing the move that the player, displaying the move's name, and giving the player a score based on the correctness of the motion and speed. F-DTW (Fast Dynamic Time Warping) algorithm has been employed in their research and other previous research [12], [13] After using F-DTW in their paper, it achieved an average accuracy of 90% and was able to recognize the movements.

In another paper, Priya & Arulselvi [14] seek to classify the poses as belonging to either the Bharatanatyam dance or Karate Kata. Their main purpose is to use a Multiview dataset to train a CNN model. Three cameras were positioned all around the participants to record movements from various angles to produce the Multiview dataset. In addition, multiple data augmentation techniques such as rotation and flipping of images were used to increase the size of the dataset. After training the model, the pose classification accuracy was 62%.

Hachaj et al., [15] discussed the use of Gesture Description Language to recognize different Karate techniques. The primary issue that this study seeks to address is the wrong classification of body joints due to the coverage by another joint in fast and complex Karate techniques. This was solved by using three kinetics to capture the skeleton from different angles and then integrating all recordings from the kinetics. This integration is done by checking if a particular body joint is not visible by one device, then it could be visible in another. After collecting data captured from the 3 kinetics, 83% of the Karate recordings were classified correctly.

Ait-Bennacer et al., [16] have used deep neural networks to classify different Karate Kata movements. Gidan Barei, Tsuki Chudan, Tsuki Jodan, Soto Uke, Shuto Uke, Mae Giri, Mawashi Giri, and Yoko Giri were the 8 movements of interest

in their study. The data set was captured from 3 views front, left, and right in order to create a multiview dataset. Lstm and ST-GCN were used to classify, give a score, and detect each of the 8 movements. The LSTM was able to recognize the movements with 96% accuracy and the ST-GCN with 91.01%.

III. METHODOLOGY

A. Dataset Collection and Labeling

We started by gathering our dataset manually from a variety of sources, including the Internet and Karate training facilities. The dataset consisted of recorded videos of Karate players performing Kata movements. This study included 9 participants of different ages and different expertise levels performing the Kata movements. Our dataset contained 4840 records, in which each record represents a single frame of the video. The videos were labeled by 2 expert Karate coaches. The coaches reviewed the videos and labeled the videos of players as either good or bad styles.

B. Data Preprocessing and Model Architecture

In order to start preprocessing our data and in turn start building our LSTM model, we first extracted pose landmarks from each frame of the video. The OpenCV library was used to loop over a specified number of frames. In order to extract the pose landmarks of the player from each frame shown in Fig. 5, the Mediapipe library was used. We stored the extracted landmarks of each frame with the proper corresponding label in a CSV file.

Since an LSTM neural network requires the input data to be in a 3-dimensional shape, we reshaped the data to be suitable as an input to the LSTM model as shown in Fig. 3 and Fig. 4.

The model's architecture consisted of 100 LSTM units as our dataset consists of a large number of features. We used 2 Dense layers which are sufficient for a binary classification task. The sigmoid function, Adam optimizer, and cross-entropy loss were used. This is because our model is primarily used for binary classification of styles to either belong to the class good or bad style.

For our training and testing splits, we used 80% of the data to train the model and used the remaining 20% for testing. The model's training process involved training the model with a number of 100 epochs. The evaluation metrics used were accuracy and loss. After the training process was complete, our LSTM Model achieved an accuracy of 90.3%.

We also used our model to classify a benchmark dataset for human pose classification, which was the MPII Human Pose Dataset [17]. Since our model is used primarily for binary classification, we only chose two poses from the MPII Human Pose Dataset. The LSTM model produced an accuracy of 100%.

IV. EXPERIMENTS AND RESULTS

A. Experiment 1

In our first experiment, we used an LSTM neural network architecture to classify each of the 20 Kata moves separately

X1	X2	X3	X4	Y
x11	x12	x13	x14	y1
x21	x22	x23	x24	y2
x31	x32	x33	x34	y3
x41	x42	x43	x44	y4
x51	x52	x53	x54	y5
x61	x62	x63	x64	y6
x71	x72	x73	x74	y7
x81	x82	x83	x84	y8
x91	x92	x93	x94	y9
...	...	...	...	...
...	...	...	...	...
...	...	...	...	...
xn1	xn2	xn3	xn4	yn

Fig. 3. An example of one dimensional input

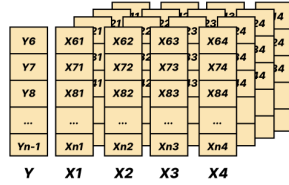


Fig. 4. Input shape for an LSTM Neural Network

using points from the 33 body joints. Each record of the data used represented joint positions with the three attributes x, y, and z for all 33 body joints in each frame of each individual movement. The attribute x represented the width, y represented the height, z represented the depth. The resulting model accuracy was 73.53%.

B. Experiment 2

In the second experiment that we conducted, we employed an LSTM model that classifies the overall performance of the player by looping over each frame of the video using the OpenCV library and extracting the x, y, and z features of the different joints from the poses' landmarks extracted using the Mediapipe library. We then used the LSTM model to classify our data. The model achieved an accuracy of 90.3%. Moreover, the model's precision, recall, and F1-Score were 91%, 90%, and 90% respectively.

V. DISCUSSION

In our paper, we used an LSTM model to classify the performance of Karate Kata players based on their body poses. We implemented two different approaches. The first one was to classify each move individually and the second one was to classify the video as a whole. The second approach produced results that were significantly improved over the first approach, and thus it was our most promising approach. To further clarify, we first loop over the frames of the video and extract the pose landmarks from each frame into a CSV file using the

OpenCV library. The Mediapipe library was used to extract the pose landmarks from each frame throughout the video. We then saved the extracted data into a CSV file. Furthermore, we reshaped our data to be suitable as an input for the LSTM network since an LSTM neural network requires the data to be in a specific shape. Finally, our model achieved 90.3% accuracy. Moreover, the model's precision, recall, and F1-Score were 91%, 90%, and 90% respectively. We compared our model's performance to that of many classical machine learning classifiers to further assess its performance. In addition, we tested our model on a benchmark dataset which was the MPII Human Pose Dataset. Our LSTM model achieved an accuracy of 100% on the benchmark dataset indicating its generalizability.

VI. CONCLUSION

In conclusion, classifying the style of Karate Kata players manually is a costly, inefficient, and time-consuming process. It can also lead to bias and unfairness. In comparison, the process of automatically classifying the style of Karate Kata players is inexpensive, efficient, and a time-saving process. In addition, classifying the style of Karate Kata players automatically ensures that all players are evaluated according to the same metrics and thus providing fairness. Moreover, classifying the style of a Karate Kata player can be done automatically using different approaches. The approach we settled on was classifying the video as a whole by looping over the frames of the video and extracting the pose landmarks from each frame since it produced the most promising results with an accuracy of 90.3%. Moreover, the model's precision, recall, and F1-Score were 91%, 90%, and 90% respectively. In addition, we also tested the model on the MPII Human Pose Dataset as an assurance of its applicability. Furthermore, we used the same approach with different classical machine learning models and in all cases, the approach of looping over the frames and extracting the pose landmarks produced the most promising results. Finally, this paper sheds light on how Deep Learning Neural Networks are efficient in different tasks. Deep Learning opens up the road to countless applications. In our case, the LSTM Neural Network proved its efficiency in the task of binary classification.

VII. FUTURE WORK

Automatically classifying the style of Karate Kata players is a branch of Human-Computer Interaction (HCI) that has not been fully scratched yet. Numerous approaches can be used as well as numerous combinations. Future work can look into novel ideas to classify the style of a Karate Kata player more efficiently. Future work can also try to work on some of the persistent problems faced during the carrying-out of the task. To further elaborate, a problem we faced during the processes of building, training, and testing the models was the fact that a larger dataset with accurate labels was needed. Future research could look into larger and more diversified datasets that are accurately labelled by multiple coaches with a wider range of Karate movements. Another approach that can be implemented

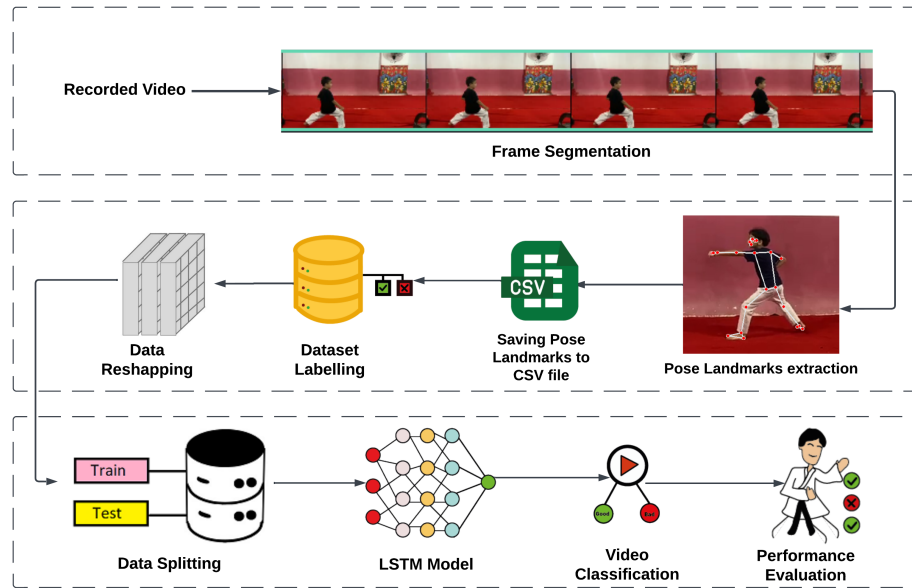


Fig. 5. System Overview

is using CNNs for feature extraction. This is because they are considered revolutionary in the task of image classification [18]. Moreover, CNNs can also be combined with other deep learning neural networks in order to improve the results, the efficiency, and the reliability of the model. Additionally, future research can focus specifically on comparing classical machine learning models with Deep Learning approaches. In addition to that, future work can also investigate the use of data augmentation techniques such as flipping and rotation of frames in videos to generate additional training data, which could help improve the model's performance. Lastly, testing the models on unseen data will ensure the real-world applicability of the model.

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